Exploratory Analysis of Philadelphia Indego Bike-Share Ridership (2018–2025)

Indego Bike-Share Ridership Analysis

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This report analyzes 7,728,268 Philadelphia Indego bike-share trips from 2018 to 2025 using Python, Pandas, Matplotlib, and Jupyter Notebook. The analysis examines station usage, weekday ridership, precipitation, temperature, and time-of-day trends. Results indicate higher ridership during midweek and warmer, drier conditions, with substantial growth in afternoon trips.

1 INTRODUCTION

Philadelphia's Indego bike-share program provides an alternative transportation option for residents and visitors. Examining ridership patterns can help identify trends in bicycle usage and the factors associated with demand.

2 DATA LOADING AND INITIAL AUDIT

Eight CSV files organized by year, were combined into a chronologically sorted DataFrame containing 7,728,268 trips and 15 columns covering trip details, stations, coordinates, bicycles, and rider passes. Missing values occurred in start_lat (726), start_lon (726), end_lat (114,050), end_lon (114,050), plan_duration (7), passholder_type (1,649), and bike_type (300,617). The remaining columns—trip_id, duration, start_time, end_time, start_station, end_station, bike_id, and trip_route_category—had no missing values.

3 STATION ANALYSIS

Station names were merged using the provided lookup file. Among matched stations, 15th & Spruce had the highest average yearly starting (18,163) and ending (17,863) rides. Table 2 lists the top 10 stations with their means and standard deviations, calculated using years with recorded trips. Approximately 13.4% of rides had no matching station names.

Table 2: Top 10 Starting and Ending Stations by Average Yearly Rides

Rank	Starting Station	Mean (SD)	Ending Station	Mean (SD)
1	15th & Spruce	18,163 (3,573)	15th & Spruce	17,863 (3,619)
2	23rd & South	14,565 (2,597)	23rd & South	14,835 (2,546)
3	Rodin Museum	13,129 (849)	Virtual Station	13,953 (5,064)
4	11th & Pine, Kahn Park	12,041 (1,950)	Rodin Museum	12,918 (873)
5	9th & Locust	11,981 (1,194)	11th & Pine, Kahn Park	12,031 (1,930)
6	11th & South	11,822 (874)	4th & Bainbridge	11,914 (1,708)
7	Philadelphia Museum of Art	11,748 (3,787)	11th & South	11,858 (754)
8	4th & Bainbridge	11,563 (1,793)	9th & Locust	11,741 (1,336)
9	17th & Locust	11,514 (4,324)	Philadelphia Museum of Art	11,568 (3,849)
10	19th & Lombard	11,353 (1,968)	17th & Locust	11,533 (4,321)

4 WEEKDAY RIDERSHIP

Mean daily rides (SD) were: Monday 2,588 (1,256), Tuesday 2,825 (1,350), Wednesday 2,848 (1,358), Thursday 2,823 (1,385), Friday 2,730 (1,357), Saturday 2,486 (1,356), and Sunday 2,213 (1,228). Wednesday had the highest average ridership, followed by Tuesday and Thursday, while Sunday had the lowest.

5 WEATHER AND TEMPERATURE ANALYSIS

Average daily rides were highest on dry days (2,942), followed by rainy (2,270) and snowy (824) days. Among 10 equal-width temperature bins, the most rides (1,656,281) occurred between 19.67°C and 24.18°C, while the fewest (5,766) occurred between -11.95°C and -7.39°C.

6 TIME-OF-DAY RIDERSHIP

Figure 1 shows increasing ridership across all four time periods from 2018–2025. Afternoon trips were the most common by 2025, while nighttime rides remained the least frequent.

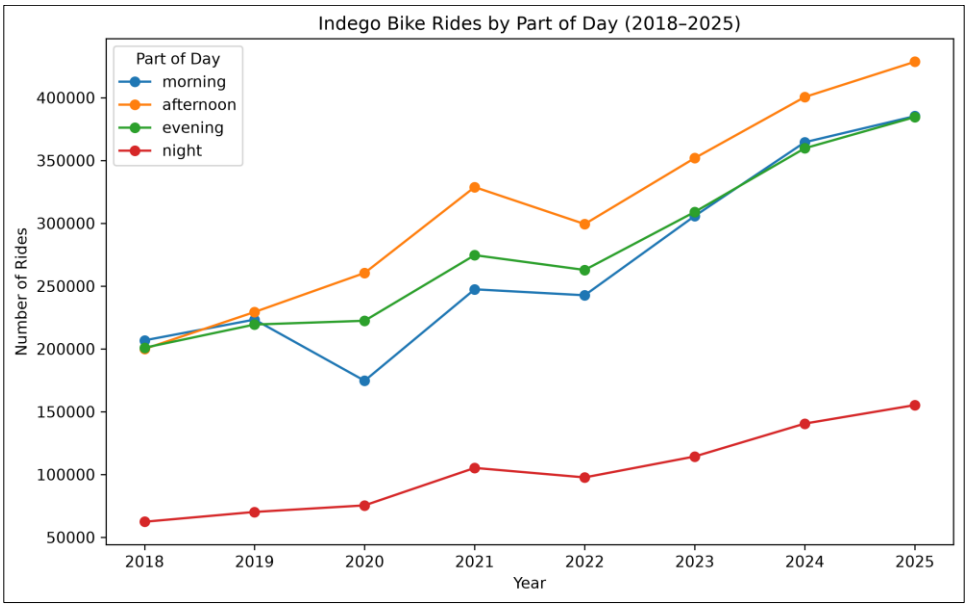


Figure 1: Annual Indego bike trips by part of day (2018–2025).

7 CONCLUSION

Indego ridership was generally highest during midweek, warmer weather, and afternoon hours. These findings highlight patterns in bike-share usage across Philadelphia. However, the analysis was limited by missing station names, incomplete weather records, and the exclusion of rides between 4:00 and 4:59 AM from the time-of-day categories.